

Model-Agnostic Adaptive Testing for Intelligent Education Systems via Meta-learned Gradient Embeddings

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The field of education has undergone a significant revolution with the advent of intelligent systems and technology, which aim to personalize the learning experience, catering to the unique needs and abilities of individual learners. In this pursuit, a fundamental challenge is designing proper test for assessing the students' cognitive status on knowledge and skills accurately and efficiently. One promising approach, referred to as *Computerized Adaptive Testing* (CAT), is to administrate computer-automated tests that alternately select the next item for each examinee and estimate their cognitive states given their responses to the selected items. Nevertheless, existing CAT systems suffer from inflexibility in item selection and ineffectiveness in cognitive state estimation, respectively. In this article, we propose a Model-Agnostic adaptive testing framework via

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Meta-learned Gradient Embeddings, MAMGE for short, improving both item selection and cognitive state estimation simultaneously. For item selection, we design a Gradient Embedding-based Item Selector (GEIS) which incorporates the concept of gradient embeddings to represent items and selects the best ones that are both informative and representative. For cognitive state estimation, we propose a Meta-learned Cognitive State Estimator (MCSE) to automatically control the estimation process by learning to learn a proper initialization and dynamically inferred updates. Both MCSE and GEIS are inherently model-agnostic, and the two modules have an ingenious connection via meta-learned gradient embeddings. Finally, extensive experiments evaluate the effectiveness and flexibility of MAMGE.

CCS Concepts: • **Applied computing** → **E-learning; Interactive learning environments**; • **Computing methodologies** → *Active learning settings*;

Additional Key Words and Phrases: Adaptive testing, intelligent tutoring system, active learning, meta-learning

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1 Introduction

The field of education has undergone a significant revolution with the advent of intelligent systems and technology [27], which aim to personalize the learning experience, catering to the unique needs and abilities of individual learners. In this pursuit, one fundamental challenge is assessing the students' cognitive status on knowledge and skills accurately and efficiently. Traditional paper-based exams, as shown in Figure 1(a), often fall short in capturing the diverse and evolving abilities of learners. To be more specific, providing all examinees with the same unpersonalized environment making it hard to guarantee the rationality and effectiveness of all the selected items [32]. For instance, if the overall difficulty of the item set is low, the more able examinees will perform similarly in the test such that we cannot precisely distinguish their states.

In light of these considerations, **computerized adaptive testing (CAT)** [32, 51] has emerged as a powerful approach in the realm of intelligent education systems. As demonstrated in Figure 1(b), CAT leverages the capabilities of intelligent algorithms to dynamically tailor the test items presented to each individual based on their responses and performance. By adaptively selecting items of appropriate difficulty levels, CAT provides a personalized and efficient assessment experience, enabling a more accurate estimation of a learner's proficiency [17]. Moreover, CAT also brings a lot of benefits including shortening test length [32], enhancing examinee engagement [33], and so on, leading to a broad range of applications in standard test organizations, such as **graduate management admission test** [38] and graduate record examinations [34].

CAT is typically constructed based on a **cognitive diagnosis model (CDM)** to estimate examinees' cognitive states from their responses. As shown in Figure 1(b), a CAT system generally comprises two main components that operate interchangeably during a test [32, 48]: (1) an *item selection* module, which recommends suitable items from a pool at each test step; and (2) a *cognitive state estimation* module, which updates the examinee's cognitive state based on their responses. This process continues until a termination criterion, such as reaching the maximum testing length, is met [49]. The final estimated cognitive state reflects the examinee's mastery of required knowledge or skills (e.g., "Function" in Math), supporting further analysis, such as user profiling and knowledge tracing [27]. Therefore, the fundamental challenge lies in establishing an effective CAT system that appropriately selects items and accurately estimates cognitive states for examinees.

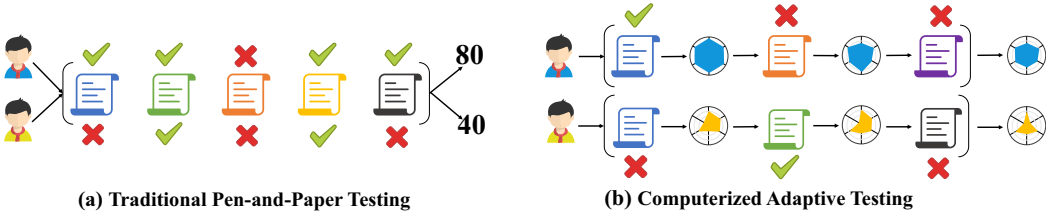


Fig. 1. (a) Traditional paper-based exams for two examinees. (b) Computerized adaptive tests comprised of alternate item selection and cognitive state estimation. The two examinees are assessed by a vanilla inadaptable estimator (upper) and an ideal proper estimator (lower), respectively.

In the realm of CAT, extensive research has concentrated on the item selection module and developed lots of *ad hoc* strategies depending on the interpretation of the underlying CDM's parameters [9]. For instance, the one-parameter logistic **item response theory (IRT)** model [12] represents the examinee ability and item difficulty with a scalar parameter, respectively. Based on this parameterization, a popular item selection strategy is to select the item with its difficulty closest to the examinee's current estimated ability [31]. According to the updated response data, the examinee's cognitive state is estimated via parameter optimization methods tied with the development of the CDM, e.g., maximum likelihood estimation [9], which is less explored in the CAT field.

Despite the significant progress made in recent years, the effectiveness and flexibility of CAT systems are still limited by some major challenges, which we summarize from two aspects. *Firstly, existing item selection strategies are mostly model-specific, resulting in inflexible CAT systems.* That is, they are deeply tied to the detail of the underlying cognitive diagnosis model and thus inapplicable to other models. Taking the above difficulty-matching strategy [31] as an instance, the method will fail if the underlying CDM has different structure and different meanings of the parameters from the IRT model. Instead, we have to seek for a specifically designed strategy for a different CDM. What's more, it is not easy to customize an item selection strategy for complex models based on a large variety of advanced techniques nowadays [29, 42, 53, 59]. Therefore, an ideal CAT framework is supposed to be model-agnostic, i.e., the CAT solution can adapt to different CDMs while keeping effective in the mean time. *Secondly, existing cognitive state estimators suffer from a lack of fast adaptability, restricting the effectiveness of CAT.* In general, currently adopted estimators start from a random initialization, and then obtain cognitive state estimation via optimization methods, such as gradient descent at each step, rigidly relying on the responses of the current examinee. Such an inadaptable estimator suffers from two disadvantages. First, simply fitting the cognitive state parameters on a small amount of data (note that adaptive tests are typically short) is prone to result in overconfidence and biased estimation. For example, when the examinee has only one correct (or wrong) response at the first test step, an ideal estimator should adapt the ability estimation to a properly higher (or lower) level but not an extreme one. Second, the estimation policy cannot be adjusted dynamically as the test steps on, leading to further limitations on the effectiveness of adaptive tests. For instance, it is reasonable to make crude estimation aggressively for fast adaptation in the initial stage while gradually making meticulous updates conservatively for fine-tuning as the test goes on. Figure 1(b) illustrates two toy cases in comparison. In the upper case, the estimator severely overestimates due to the examinee's first correct response and thus provides extremely difficult items in the following interactions, resulting in poor estimation **accuracy (ACC)** and examinee's experience. In the lower one, an ideal estimator starts with a proper estimation and smoothly adjusts to a refined result.

In our preliminary work [8], we tackled the first challenge with our proposed **model-agnostic adaptive testing (MAAT)** framework. By incorporating the idea of active learning [41], we designed a novel item selection heuristic based on the **expected model change (EMC)**. Specifically, an item with a large expected gradient is considered to be informative, because it indicates a potentially big change to the cognitive state estimation of the underlying CDM once we know the examinee's response to this item. Besides, to prevent CAT from selecting out-of-distribution items, we associate each item with an **importance weighted knowledge coverage (IWKC)** indicating how much it represents the whole item pool. Combining EMC and IWKC together, MAAT selects the next item that is both informative and representative. Note that the above procedure hides the detail of the underlying CDM and is thus model-agnostic.

In this article, we further extend our previous work to support a **MAAT framework for intelligent education systems via meta-learned gradient embeddings**, MAMGE for short, which is effective and flexible in both item selection and cognitive state estimation. To address the model-specific issue in item selection, we design a **gradient embedding-based item selector (GEIS)** along the lines of MAAT [8]. Specifically, we represent each item with a gradient embedding that can be efficiently computed on any differentiable CDM. Then, we quantify the informativeness and representativeness of each item via the length and direction of gradient embeddings, respectively, and naturally design a pair of model-agnostic item selection heuristics. To achieve fast adaptability for cognitive state estimation, inspired by the ideology of meta-learning, we propose a **meta-learned cognitive state estimator (MCSE)** that intelligently learns how to estimate better in a data-driven manner. Principally, MCSE automatically decides all the elements of the cognitive state estimation procedure (including initialization, update direction, and step size) via the meta-experiences from historical examinees. What's more, these elements are dynamically inferred along with the testing process of the current examinee. Finally, by refining the original gradient embeddings used in GEIS with meta-learned ones, we show that the cognitive state estimation module MCSE and the item selection module GEIS can be well connected, which further improves the effectiveness of the CAT system.

In summary, by extending the consideration of CAT systems from solely item selection strategies to the complete selection-estimation procedure, we further address the challenge of achieving an effective model-agnostic CAT system with our proposed framework MAMGE, including a GEIS and an MCSE. Finally, we conduct extensive evaluation on four real-world datasets with two base CDMs and show that MAMGE outperforms the competing baselines, revealing its effectiveness and flexibility.

The rest of the article is organized as follows. Section 2 covers the research fields as well as the previous work related to our article. In Section 3, we formally define the studied adaptive testing problem. Section 4 introduces the design of the proposed MAMGE framework, which firstly provides an overview (Section 4.1), then goes into the detail of the two specific modules, i.e., GEIS (Section 4.2) and MCSE (Section 4.3), and finally gives a discussion on the methodology (Section 4.4). In Section 5, we show the experimental analysis for our framework. At last, we summarize the presented work and look forward to future work in Section 6.

2 Background and Related Work

2.1 Adaptive Testing

Adaptive testing has been recognized as an increasingly important assessment mode due to its outstanding efficiency as well as ACC [9]. Previous researchers mainly focus on designing designated item selection strategies for different CDMs [28], among which the representative is the IRT model [12]. Lord et al. [31] proposed to select the item with the maximum Fisher item information,

which has been the most popular algorithm over decades; Chang et al. [10] utilized the global **Kullback–Leibler index (KLI)** rather than pointwise information to improve robustness. These methods are later on extended for **multi-dimensional IRT (MIRT)** [19, 52]. For other types of CDMs, designated strategies are developed [37, 49, 61]. However, these model-specific strategies lack flexibility and can hardly extend to new CDMs, especially the recently popular deep learning models [29, 53]. Therefore, some researchers turn to seeking for a model-agnostic framework for adaptive testing. Viewing the problem from an active learning perspective, Bi et al. [8] proposed a universally adaptable heuristic for item selection. Another line is to directly learn a strategy from data. Nurakhmetov et al. [36] applied reinforcement learning to train an agent; Ghosh et al. [16] tried to obtain the optimal strategy with bilevel optimization. In comparison, heuristic-based methods ignore the rich information in data, while data-driven methods are difficult to train and prone to suffer from biases and scarcity of data. Though making great progress in selecting proper items, the previous researches yet pay little attention to the cognitive state estimation part for CAT. Once receiving the response from the examinee, existing CAT systems minimize the loss function (e.g., cross entropy loss) with typical optimization methods (e.g., gradient descent). However, such straightforward estimation tends to overfit because of the inherently short length of CAT. Ghosh et al. [16] alleviated this problem through a state-of-the-art end-to-end CAT solution **bilevel optimization-based CAT (BOBCAT)**, which yields a proper *initialization* for cognitive state parameters via a novel bilevel-optimization framework. Nevertheless, BOBCAT still lacks consideration toward a proper parameter *update* policy to achieve fast adaptability and avoid overfitting during testing. Besides, there are also recent works dealing with some other topics on the CAT problem, such as robustness [62] and search efficiency [18], which is orthogonal to our work.

2.2 Active Learning

Active learning [41] is a machine learning paradigm aiming to maximize the performance of models by selecting minimum labeled samples, which has been well adapted to the development of intelligent systems [7, 30]. Generally, there are two major perspectives for sample selection in active learning, i.e., informativeness and representativeness. From the view of sample informativeness, it is suggested to select those samples which are uncertain to the current model. Different uncertainty measurements lead to a variety of informativeness-based sampling methods, such as decision boundary based method [6] and Bayesian method [14]. Informativeness can also be measured with model influence, which is more relevant to our work. In this kind of methods, the samples are considered uncertain if they have a great impact on the model parameters. Such impact can be analyzed via influence function [47] or expected gradient length [21, 58]. From another point of view, rather than the information in each sample, we can select samples which can well represent the overall sample distribution. Intuitively, training on a few representative samples can approximate the effect of training on more samples of the underlying distribution and can also prohibit the model from out-of-distribution samples. Commonly used representativeness measurements include the geometrical structure of data and the relationship among neighbors, and so on. For example, Sener et al. [40] sought for the global representatives of data with core-set, while Yang et al. [56] formulated representativeness maximization as a set cover problem. Since the two perspectives provide different kinds of heuristics, it is natural to take both informativeness and representativeness into consideration [2, 55]. Besides, there are also some data-driven approaches which automatically learn heuristics from data [3, 24]. In our previous work [8], we draw a parallel between adaptive testing and active learning to design a pure heuristic-based item selection strategy. However, in this article, we extend to a hybrid approach utilizing meta-learned data information as well, which will be discussed in detail in Section 4.4.

2.3 Meta-Learning

Meta-learning, also called learning-to-learn, studies how differently machine learning approaches perform on a wide range of learning tasks, and how to learn new tasks faster by learning from this meta-experience [20]. According to the way of encoding meta-knowledge, previous work can be classified into three types: model-based, metric-based, and optimization-based. Model-based meta-learning directly utilizes a black-box meta-network to store meta-knowledge through memory-augmented architecture and control the base learner [35, 39], which suffers from steep training difficulty. Metric-based meta-learning embeds meta-knowledge into a metric space for objects so that classification tasks can be more efficient [23, 44, 46, 50], but the application scenarios are relatively limited. Optimization-based meta-learning injects meta-knowledge into the optimization algorithm of the base learner to enable fast adaptation to a few samples, specifically focusing on gradient back-propagation optimization due to its dominance nowadays. The most popular methods are the model-agnostic meta-learning framework [13] and its variants [1, 5, 20, 22], which learn a good parameter initialization for models via meta-gradient descent. Furthermore, recent researches have revealed and explored the importance of other optimization components besides initialization, e.g., learning rate [4, 26, 63]. The cognitive state estimation module in our proposed framework falls into the optimization-based genre. We regard conducting adaptive testing for each examinee as an individual task and design a MCSE that adapts fast for new examinees by controlling all of the initialization, update direction, and step size.

3 Preliminaries and Problem Definition

This section introduces necessary terminologies and definitions. We firstly define the CDM in an abstract form capturing common characteristics of most CDMs, making our discussion inherently model-agnostic. Then, we introduce the general adaptive testing system based on the abstract CDM, including what components it consists of and how it conducts tests for examinees. Finally, we formally define the problem studied in this article.

Suppose there are m examinees $\mathcal{E} = \{1, \dots, m\}$ and n items $\mathcal{I} = \{1, \dots, n\}$. We denote the response of examinee e on answering item i as $r_{ei} \in \{1, 0\}$ (i.e., right or wrong). A CDM $\mathcal{M}$ is defined as a parameterized function $\mathcal{M}(\theta, \varphi; \omega)$, where θ denotes the examinee's cognitive state (e.g., ability level), φ denotes the item feature (e.g., difficulty level), and $\mathcal{M}(\cdot, \cdot; \omega)$ is a link function with possible trainable parameters ω (e.g., **neural networks (NNs)**). For simplicity, we omit the parameters of the link function in the rest of this article and denote a CDM as $\mathcal{M}(\theta, \varphi)$. Given an examinee $e \in \mathcal{E}$ and an item $i \in \mathcal{I}$, we can predict the response by $\hat{r}_{ei} = \mathcal{M}(\theta_e, \varphi_i)$, $\hat{r}_{ei} \in [0, 1]$. For example, the most commonly used CDM, IRT [52], has its simplest form as follows:

$$\hat{r}_{ei} = \mathcal{M}(\theta_e, \varphi_i) = \sigma(\theta_e - \varphi_i), \quad (1)$$

where $\theta_e \in \mathbb{R}$ and $\varphi_i \in \mathbb{R}$ are scalar cognitive state parameters representing the examinee's ability level and the item's difficulty, respectively, and $\sigma(\cdot)$ is the sigmoid function. Typically, the item features $\varphi_i (i \in \mathcal{I})$ and the parameters ω are learned from massive historical data and then fixed for future new examinees.

Given the base CDM $\mathcal{M}(\theta, \varphi)$, an adaptive testing system comprises of two fundamental modules: item selection module and cognitive state estimation module. For an examinee $e \in \mathcal{E}$, the two modules present the adaptive test in an iterative fashion by working alternately, where each step consists of the following two parts: (1) *Selection*: The item selection module selects a personalized item i_t according to the estimated cognitive state θ_e^{t-1} for the examinee, and then obtains the response r_{ei_t} from the examinee; (2) *Estimation*: The cognitive state estimation module updates θ_e^{t-1}

Algorithm 1: MAMGE

Input: cognitive diagnosis model $\mathcal{M}(\theta, \varphi)$; examinee set $\mathcal{E} = \mathcal{E}_{train} \cup \mathcal{E}_{test}$; item set $\mathcal{I}$; test length $T \in \mathbb{Z}_+$
Output: final cognitive state estimation θ_e^T
Meta-optimization for MCSE: train meta-parameters of MCSE on $\mathcal{E}_{train}$ (detail in Section 4.3)
for each examinee $e \in \mathcal{E}_{test}$ **do**
 initial estimation θ_e^0
 set selected and unselected item set $\mathcal{I}_S^0 = \emptyset, \mathcal{I}_U^0 = \mathcal{I}$
 for $t = 1, \dots, T$ **do**
 Selection with GEIS: select the item $i_t \in \mathcal{I}_U$ with gradient embedding based heuristics (Section 4.2)
 Estimation with MCSE: update the cognitive state $\theta_e^{t-1} \rightarrow \theta_e^t$ by meta-learner (Section 4.3)
 $\mathcal{I}_U^t = \mathcal{I}_U^{t-1} \setminus \{i_t\}, \mathcal{I}_S^t = \mathcal{I}_S^{t-1} \cup \{i_t\}$
 end for
end for

to θ_e^t according to the examinee's response r_{ei_t} . The final cognitive state estimation θ_e^T is output as the test result.

In practice, the examinees can be classified as $\mathcal{E} = \mathcal{E}_{train} \cup \mathcal{E}_{test}$. $\mathcal{E}_{train}$ denotes the historical examinees that have recorded responses used to train the base CDM (i.e., calibrate the item features and link function parameters) as well as the adaptive testing system if needed. $\mathcal{E}_{test}$ denotes those new examinees supposed to be totally unknown. Basically, the aim of our study is to estimate the cognitive state as accurately as possible. Therefore, for an examinee $e \in \mathcal{E}$, it is necessary to quantitatively evaluate how accurate the output θ_e^T is. Since it is impractical to obtain the true value for θ_e , we seek for a surrogate task to evaluate θ_e^T . Specifically, we predict the examinee's future response based on θ_e^T (e.g., on a validation item set) and assume that the more accurate the prediction is, the more accurate θ_e^T is, too. In this way, we can leverage typical evaluation protocols for prediction tasks such as **area under the curve (AUC)** (Section 5.1.1). Generally, we refer to such a protocol as the *testing effectiveness* and denote it as $\psi(\theta_e^T)$. Therefore, our studied adaptive testing problem can be formulated as follows.

Problem definition. Given a base CDM $\mathcal{M}(\theta, \varphi)$ and training data from historical examinees $\mathcal{E}_{train}$, our goal is to realize an effective adaptive testing system such that, for a new examinee $e \in \mathcal{E}_{test}$, the testing effectiveness $\psi(\theta_e^T)$ is maximized.

4 MAMGE

4.1 Framework Overview

To realize an effective adaptive testing system, we propose an MAAT framework via Meta-learned Gradient Embeddings, namely MAMGE for short. As summarized in Algorithm 1, for any examinee e , at each test step $t = 1, \dots, T$, the underlying CDM has a temporary cognitive state estimation from the last step, θ_e^{t-1} . Based on the estimation, the item selection module, named as GEIS, selects the next item i_t for the examinee by associating the candidate items with so-called gradient embeddings and ranking their priorities with carefully designed heuristics. Then, the examinee receives the item i_t and gives his/her response r_{ei_t} . According to the response, the cognitive state estimation module, named MCSE, automatically updates θ_e^{t-1} to θ_e^t via a meta-learning based approach. Figure 2 illustrates the overall flowchart. In the rest of this section, we will introduce the two modules in more detail, respectively.

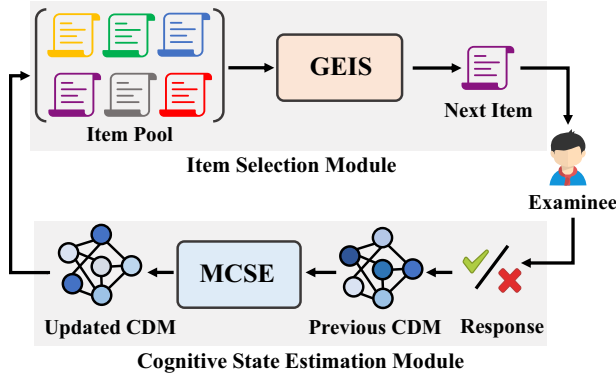


Fig. 2. Overview of our MAMGE. The adaptive test workflow for an examinee consists of item selection and cognitive state estimation, with the proposed GEIS (detail in Figure 3) and MCSE (detail in Figure 4) lying in, respectively.

4.2 GEIS

In a CAT system, the item selection module aims at selecting a personalized item for the current examinee so that his cognitive state can be better estimated at the next step. Although a variety of selection strategies have been proposed [9], an inherent issue lies in their lack of model-agnostic property. To make the item selector flexible to different CDMs instead of designated to specific one, we leverage the ideology of active learning [41]. Specifically, we treat each item as a data sample, upon which the response of the examinee corresponds to the label (correct or wrong). Similarly, the CDM plays the role of a predictive model. From this perspective, item selection in CAT can be transformed into a standard active learning problem whose key lies in quantifying the value of unlabeled samples to maximize model performance. There are two aspects worth considering. On the one hand, the *informativeness* of each item is the most important consideration, which ensures utility of the selected items, i.e., providing enough information for the CDM about the examinee's cognitive state. On the other hand, the *representativeness* of each item w.r.t. the whole item pool should also be taken into account, which ensures that the test is robust and comprehensive. Along this line, we propose a novel GEIS. As illustrated in Figure 3, we firstly introduce the concept of *gradient embedding* to measure the above two kinds of value associated with each item in a unified manner. Then, we select the best item based on a pair of well-designed heuristics.

4.2.1 The Power of Gradient. To measure the value of an item, we consider how it influences the CDM. Principally, once we obtain the examinee's response to an item, the CDM updates its cognitive state estimation, typically by computing the gradient w.r.t. a loss function and then performing gradient-based optimization. In this sense, the gradient can be recognized as a representation of the item's test-related feature. Specifically, we argue that: the *length* and *direction* of the gradient convey information about the informativeness and the representativeness of each item, respectively.

Generally, the length of a gradient implies how much the CDM will update its cognitive state estimation (i.e., θ) if knowing the examinee's response to the item. With a large gradient length, θ will probably have a great change, therefore the item conveys a nice bit of information about the examinee's cognitive state. Conversely, with a small gradient length, the item will likely have little impact on θ as if the examinee had not answered this item, which can hardly help the CDM diagnose the examinee better. In summary, we consider the item to be informative if it has a great influence on the CDM, and its informativeness can be represented by the gradient length.

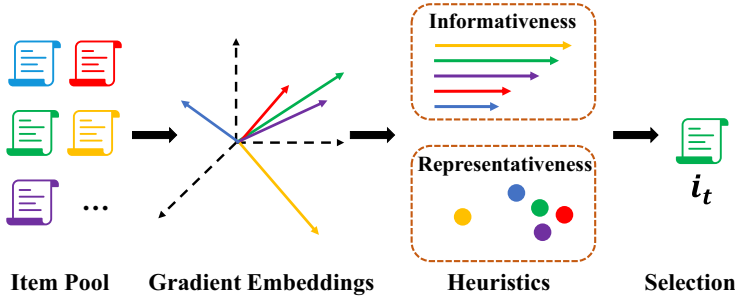


Fig. 3. Illustration of the GEIS. At each step during the adaptive test, every item in the item pool is represented with a gradient embedding. Then, the embeddings are used to calculate two types of heuristics which quantify the value of items. Finally, the final selection is determined according to the heuristics.

In addition to the length, the direction of a gradient captures information on how the item affects the CDM's estimation as well. For a CDM where cognitive state is represented in multiple dimensions, different dimension typically has different intrinsic effects. For instance, MIRT [52] supposes each dimension to be the examinee's ability level on a specific knowledge concept (e.g., geometry and algebra); deep learning models [53] learn complex fusion features from data. Correspondingly, two items are considered to have similar effects on cognitive state estimation if their gradient embeddings have similar directions, and vice versa. By extension, the gradient directions of all the items can convey information about the underlying distribution of the whole item pool. We utilize such information to optimize the representativeness of the selected items, i.e., select the items whose gradient direction is more similar to the underlying distribution. On the one side, a representative item contributes more to the test, because taking this item has the same effect as taking more similar items in a sense. On the other side, we can avoid selecting out-of-distribution items so that the test can be more robust.

Therefore, the gradient of an item can be utilized to measure its value during item selection. However, it is difficult in practice to obtain the response data on unselected items. Instead, we compute the gradient with the CDM's current prediction on the item as proxy labels. In this way, we can associate each unselected item with the gradient which reflects certain test-related feature (i.e., informativeness and representativeness), thus we refer to it as *gradient embedding*.

4.2.2 Gradient Embedding-Based Heuristics. Based on the concept of gradient embedding discussed above, we introduce two types of heuristics for item selection which considers informativeness and representativeness via gradient length and gradient direction correspondingly. For this moment, we formally denote the gradient embedding of the item $i \in \mathcal{I}$ at the t -step when testing an examinee $e \in \mathcal{E}$

$$\mathbf{g}_{ei}^t = \nabla_{\theta} l(\mathcal{M}(\theta_e^{t-1}, \varphi_i), \mathbb{I}(\mathcal{M}(\theta_e^{t-1}, \varphi_i) > 0.5)), \quad (2)$$

which is a proxy gradient w.r.t. a loss function $l(\cdot, \cdot)$ (e.g., binary cross entropy loss) that is calculated with the predicted response $\mathcal{M}(\theta_e^t, \varphi_i; \omega)$ and the pseudo label given by the indicator function $\mathbb{I}(\cdot)$. For simplicity, we omit the scripts t (denoting the step) and e (denoting the examinee) and use $\mathbf{g}_i$ for the gradient embedding of item i without ambiguity. After introducing the MCSE module, we will refine the formulation of $\mathbf{g}_i$ with a meta-learned gradient (Section 4.3.3) while the other parts of the item selection module remain the same.

To quantify the informativeness for each individual item $i \in \mathcal{I}$, we define the following informativeness based heuristic via the length of its gradient embedding

$$\text{Info}(i) = \|\mathbf{g}_i\|, \quad (3)$$

where $\|\cdot\|$ denotes the Euclidean norm function.

To capture the information about representativeness, i.e., to what extent an item can represent all the unselected items, we utilize the direction of gradient embeddings. Specifically, we measure the representativeness of an item by its cosine similarity with the other unselected items, and naturally define the following representativeness based heuristic:

$$\text{Repr}(i; \mathcal{I}_U) = \frac{1}{|\mathcal{I}_U|} \sum_{j \in \mathcal{I}_U} \frac{\mathbf{g}_i \cdot \mathbf{g}_j}{\|\mathbf{g}_i\| \|\mathbf{g}_j\|}. \quad (4)$$

4.2.3 Two-Stage Item Selection. Given the heuristics defined above, we now introduce the complete strategy used in GEIS, which selects each item in a two-stage manner. Specifically, at step t , a candidate item set $\mathcal{I}_C$ is selected from the unselected items $\mathcal{I}_U$, which consists of K_C items with maximum informativeness based heuristic. Then, one final item with the maximum representativeness based heuristic is selected from $\mathcal{I}_C$

$$\begin{aligned} i_t &= \arg \max_{j \in \mathcal{I}_C} \text{Repr}(j; \mathcal{I}_U^{t-1}), \\ \text{s.t. } \mathcal{I}_C &= \arg \max_{A \subset \mathcal{I}_U^{t-1}, |A| \leq K_C} \sum_{i \in A} \text{Info}(i). \end{aligned} \quad (5)$$

In comparison to directly combining the two heuristics (e.g., by weighted sum or product), conducting two sequential stages as above have two advantages. First, it is more computationally efficient because we need only compute the representativeness based heuristic for a small number of items in the second stage. Second, we empirically find that the hyperparameter (i.e., K_C) is easier to tune than those of other combination methods.

4.3 MCSE

While the proposed GEIS achieves model-agnostic item selection for CAT, yet we lack proper cognitive state estimation after the examinee responses to our carefully selected items. Most of the existing CDMs, especially the recent deep learning based models, use gradient-based optimization methods such as stochastic gradient descent for cognitive state estimation. Commonly, there are three key ingredients in defining an estimator (i.e., an optimizer): initialization, update direction, and step size. Existing CDMs typically realize the estimator in a simple way: (1) the cognitive state parameters are randomly initialized from some predefined distribution such as a uniform distribution; (2) given the examinee's responses, the update direction is determined by the gradient of a cost function such as the cross entropy loss; and (3) the step size is usually manually set as a hyperparameter, usually with an optional decayed inner loop to optimize multiple iterations until convergence.

The above vanilla inadapative estimation method suffers from the following two drawbacks. First, the estimator is prone to be overconfident in the examinee's existing responses and thus produce extreme values in estimation. For instance, if the examinee answers the first question correctly, the estimator tends to overestimate the examinee's ability, which results in selecting a far more difficult item in the following step. In turn, if the examinee performs badly later, the ability of this examinee is likely to be underestimated. Overall, the testing process appears to be unstable. Second, a vanilla estimator is not able to adjust its estimation policy during the test.

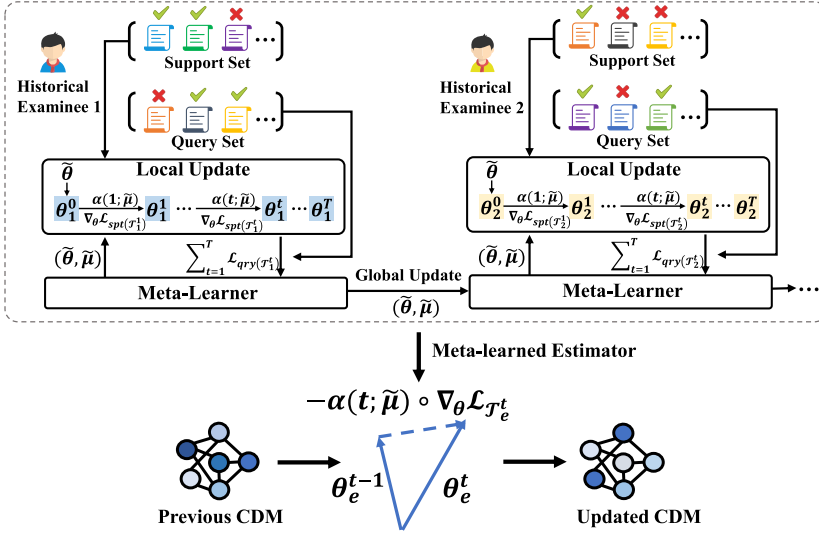


Fig. 4. Illustration of the MCSE. The upper part demonstrates the bilevel meta-optimization procedure with historical examinees corresponding to Algorithm 2 except that the local update is illustrated with a single examinee rather than a batch for visual clarity. The lower part shows how the meta-learned estimator updates the cognitive state parameters for a new examinee corresponding to Equation (8).

For example, in the initial stage of the test, it is urgently needed to seek fast for an approximate range of the examinee's cognitive state, so coarse-grained estimation policies, such as a relatively large step size, are required. In contrast, as the uncertainty of the cognitive state gradually decreases with the progress of the test, it would be better to fine-tune for a more precise estimation.

The key to tackle the above problems is to rely the estimation not only on a small number of data from the in-test examinee but also the massive data from all the historical examinees. Before directly learning the cognitive state from the current examinee, can the CAT system *learn how to learn* via historical examinees? This challenge can be naturally turned into a problem of learning-to-learn, also known as meta-learning. Therefore, we propose an MCSE that properly updates the examinee's cognitive state with only a few responses by learning to learn from massive historical records, which is shown in Figure 4. Specifically, we firstly formulate cognitive state estimation as a meta-learning problem and design a sophisticated meta-learner. Then, we recognize each historical examinee as a meta-training task and leverage meta-optimization to train the meta-parameters on historical data to facilitate estimation for new tasks. More detail will be discussed in this subsection, including the structure of the meta-learner and the procedure of meta-optimization. As last, we refine GEIS with meta-learned gradient embeddings produced by MCSE to link up item selection and cognitive state estimation within a unified framework.

4.3.1 Meta-Learner. Rather than updating the cognitive state parameters iteratively from random initialization using gradient descent with manually designed step size as vanilla estimators, the basic idea behind MCSE is to automatically determine all the ingredients of the estimator (i.e., initialization, update direction, and step size) and adaptively adjust them during testing. Mathematically, at the t th step of the test for the examinee e ($t = 1, \dots, T$), the estimation step of a typical

vanilla estimator can be formulated as follows:¹

$$\begin{cases} \theta_e^0 \sim p(\theta) \\ \theta_e^t = \theta_e^{t-1} - \alpha \nabla \mathcal{L}_{\mathcal{T}_e^t}(\theta_e^{t-1}) \end{cases} \quad (6)$$

At the beginning of the test, initial cognitive state parameters θ_e^0 is sampled from a prior distribution $p(\theta)$. Then at step t , the parameters θ_e^{t-1} is updated via gradient descent with a scalar step size α . $\mathcal{T}_e^t = \{(i_{t'}, r_{ei_{t'}})\}_{t'=1}^t$ represents the responses of the examinee so far, and $\mathcal{L}_{\mathcal{T}_e^t}(\theta_e^{t-1})$ is the empirical loss

$$\mathcal{L}_{\mathcal{T}_e^t}(\theta_e^{t-1}) = \frac{1}{t} \sum_{t'=1}^t l(\mathcal{M}(\theta_e^{t-1}, \varphi_{i_{t'}}), r_{ei_{t'}}), \quad (7)$$

with a loss function $l(\cdot, \cdot)$ such as the binary cross entropy loss. Correspondingly, $\nabla \mathcal{L}_{\mathcal{T}_e^t}(\theta_e^{t-1})$ is the gradient of $\mathcal{L}_{\mathcal{T}_e^t}(\theta_e^{t-1})$ w.r.t. the cognitive state parameters.

In contrast, we present the following meta-learner in a similar form but with two meta-learned terms instead:

$$\begin{cases} \theta_e^0 = \tilde{\theta} \\ \theta_e^t = \theta_e^{t-1} - \alpha(t; \tilde{\mu}) \circ \nabla \mathcal{L}_{\mathcal{T}_e^t}(\theta_e^{t-1}) \end{cases} \quad (8)$$

which alters Equation (6) with two key differences: (1) rather than randomly draw the initial parameters, we initialize the estimator for each new examinee with an *initialization term* $\tilde{\theta}$; and (2) in contrast to following the original gradient with a manually set step size, we dynamically determine both the update direction and the step size at each test step t by modifying the gradient with an *adaptation term* $\alpha(t; \tilde{\mu})$, where $\circ$ denotes element-wise product. We use the tilde marks to indicate that $\tilde{\theta}$ and $\tilde{\mu}$ are *meta*-parameters to be trained. We first discuss the basic structure and motivation of the meta-parameters in the rest of this subsection. Then, we introduce the detailed meta-optimization procedure. In addition, an intuitive metaphor-based interpretation of the meta-parameters is illustrated in Section 4.4.

(1) *Initialization term.* $\tilde{\theta}$ has exactly the same structure as the cognitive state parameters of the CDM. The idea behind is to find the optimal initialization for task-specific fine-tuning by taking the overall task distribution into consideration. Take the IRT model [52] as an example where each examinee is represented with a scalar ability level. We can intuitively consider the parameter estimation as a search problem. If we wish to find the true value fast, similar to an efficient binary search, it would be better to start from an intermediate value w.r.t. the overall ability distribution rather than an extreme one. In this way, the term $\tilde{\theta}$ helps adapt fast to the new examinee's cognitive state via an appropriate initialization meta-learned from historical data.

(2) *Adaptation term.* Different from $\tilde{\theta}$, $\tilde{\mu}$ contains a set of meta-parameters with a more complex structure behind

$$\alpha(t; \tilde{\mu}) = \text{ReLU}(\alpha_{base} + \gamma \alpha_{dyn}^t), \quad (9)$$

$$\alpha_{dyn}^t = \tanh(W h_t), \quad (10)$$

$$[h_t, c_t] = \text{LSTM}(\nabla \mathcal{L}_{\mathcal{T}_e^t}(\theta_e^{t-1}), h_{t-1}, c_{t-1}; \Psi). \quad (11)$$

As illustrated above, $\alpha(t; \tilde{\mu})$ is composed of (1) a base part α_{base} that keeps constant during the whole test and (2) an additional dynamic part α_{dyn}^t that is dynamically inferred during testing. The **rectified linear unit (ReLU)**($\cdot$) regularizes $\alpha(t; \tilde{\mu})$ to be nonnegative, and $\gamma \in (0, +\infty)$ is

¹The procedure is simplified in purpose of clarity. For example, we omit the inner-loop iteration in multi-step gradient descent and the possible variation of α in advanced learning rate decaying policies.

a hyper-parameter that provides optional weight for α_{dyn}^t . In this way, the meta-learner can dynamically adjust the policy along with the on-going test. To infer the dynamic part, we leverage a **Long short-term memory (LSTM)** network parameterized by Ψ (Equation (11)). The ordinary gradient at each step, $\nabla \mathcal{L}_{\mathcal{T}_e^t}$, is supposed to contain the information about parameter estimation and thus serves as the input into the network. For example, gradient with a larger norm might indicate more uncertainty at the current step. h_t and c_t are the hidden states and the cell states, respectively, both of which are zero-initialized. The final output is transformed from the current hidden state and with a hyperbolic tangent activation function as shown in Equation (10). To sum up, the meta-learned adaptation term $\tilde{\mu}$ consists of three parts of meta-parameters: α_{base} (Equation (9)), W (Equation (10)), and Ψ (Equation (11)).

In summary, for an examinee e , our meta-learner first initializes the cognitive state parameters θ_e^0 with $\tilde{\theta}$. Then at step t of the test, given a small number of testing responses $\mathcal{T}_e^t = \{(i_t, r_{ei_t})\}_{t'=1}^t$, the meta-learner updates θ_e^{t-1} to θ_e^t with adaptively inferred step size and update direction which are implicitly implemented in $\alpha(t; \tilde{\mu}) \circ \nabla \mathcal{L}_{\mathcal{T}_e^t}(\theta_e^{t-1})$. It is also worth noting that, at each test step t , we optimize the cognitive state parameters with strictly a single gradient descent rather than potentially multiply rounds in vanilla estimators. As will be discussed later (Section 4.4), it plays a role in linking up item selection and cognitive state estimation.

4.3.2 Meta-Optimization. We now turn to the meta-optimization procedure for the meta-learner, i.e., to learn the initialization term $\tilde{\theta}$ and the adaptation term $\tilde{\mu}$ (Equation (8)) so that the cognitive state θ_e^t can be estimated well for any new examinee e . From the perspective of meta-learning, we regard conducting adaptive testing for any examinee e as a task $\mathcal{T}_e$ and consider the distribution over tasks $p(\mathcal{T})$ which both historical and up-coming examinees are drawn from. Along this idea, we can train the meta-parameters on historical data to achieve fast adaptation on new data for the meta-learner. As shown in Figure 4, each meta-training task $\mathcal{T}_e$ consists of a support set $spt(\mathcal{T}_e)$ and a query set $qry(\mathcal{T}_e)$. In our context, $spt(\mathcal{T}_e)$ is composed of the examinee's responses to all the items which are selected during the whole adaptive test, and $spt(\mathcal{T}_e)^t$ with a superscript t contains those items selected during the first t steps of the test. Accordingly, $qry(\mathcal{T}_e)$ contains the examinee's responses to a set of target items which are held out in advance for meta-validation. Mathematically, the training objective of our meta-learner can be formulated as follows

$$\min_{\tilde{\theta}, \tilde{\mu}} \mathbb{E}_{\mathcal{T}_e \sim p(\mathcal{T})} \left[\sum_{t=1}^T \mathcal{L}_{qry(\mathcal{T}_e)}(\theta_e^t) \right], \quad (12)$$

where θ_e^t is iteratively calculated with dependence on meta-parameters $\tilde{\theta}$ and $\tilde{\mu}$, similar to Equation (8)

$$\begin{cases} \theta_e^0 = \tilde{\theta} \\ \theta_e^t = \theta_e^{t-1} - \alpha(t; \tilde{\mu}) \circ \nabla \mathcal{L}_{spt(\mathcal{T}_e)^t}(\theta_e^{t-1}). \end{cases} \quad (13)$$

Note that we sum up the accumulated loss at each step t to optimize the cognitive state estimation for the whole adaptive test. Algorithm 2 shows the detailed training procedure of our meta-learner. Principally, we iteratively sample a mini-batch of tasks from the distribution $\mathcal{T}_e \sim p(\mathcal{T})$ by equivalently sampling from historical examinees $e \in \mathcal{E}_{train}$. For each meta-training task $\mathcal{T}_e$, we make up the support set by randomly selecting a sequence of logs $spt(\mathcal{T}_e) = \{(i_t, r_{ei_t})\}_{t=1}^T$, where T is the adaptive testing length. For the query set, rather than hold out a small number of target items, we find it more robust to use *all* the examinee's response records on the item pool, i.e., $qry(\mathcal{T}_e) = \{(i, r_{ei})\}_{i \in \mathcal{I}}$. Starting from random initial meta-parameters $\tilde{\theta}$ and $\tilde{\mu}$, the algorithm performs *local updates* by calculating the training loss and adjusting the parameters on each support

Algorithm 2: The Meta-Optimization Procedure for (MCSE)

Input: learning rate for meta-parameters β
Output: meta-parameters $\tilde{\theta}, \tilde{\mu}$
 randomly initialize $\tilde{\theta}, \tilde{\mu}$
while not converge **do**
 sample a mini-batch of examinees $\mathcal{B}$ from $\mathcal{E}_{train}$
 for each examinee $e \in \mathcal{B}$ **do**
 set local parameter $\theta_e^0 = \tilde{\theta}$
 for $t = 1, \dots, T$ **do**
 Local Update: $\theta_e^t \leftarrow \theta_e^{t-1} - \alpha(t; \mu) \circ \mathcal{L}_{spt(\mathcal{T}_e)^t}(\theta_e^{t-1})$
 end for
 end for
 Global Update:
 $\tilde{\theta} \leftarrow \tilde{\theta} - \beta \sum_{e \in \mathcal{B}} \nabla_{\tilde{\theta}} \sum_{t=1}^T \mathcal{L}_{qry(\mathcal{T}_e)}(\theta_e^t)$
 $\tilde{\mu} \leftarrow \tilde{\mu} - \beta \sum_{e \in \mathcal{B}} \nabla_{\tilde{\mu}} \sum_{t=1}^T \mathcal{L}_{qry(\mathcal{T}_e)}(\theta_e^t)$
end while

set and then performs a *global update* by minimizing the test loss with the adapted meta-parameters on the query sets. The bilevel iteration repeats until the meta-parameters converge.

4.3.3 Refining Item Selection with Meta-Learned Gradients. With the proposed MCSE module, we can properly update the cognitive state estimation by a single-step meta-learned gradient. It is worth recalling that the basic idea of item selection in GEIS is to measure the value of an item via its gradient embedding. Along this line, we can refine the gradient embedding in GEIS (Equation (2)) as follows

$$\begin{aligned} g_{ei}^t &= \alpha(t; \tilde{\mu}) \circ \nabla_{\theta} \hat{\mathcal{L}}_i(\theta_e^{t-1}), \\ \hat{\mathcal{L}}_i(\theta_e^{t-1}) &= l(\mathcal{M}(\theta_e^{t-1}, \varphi_i), \mathbb{I}(\mathcal{M}(\theta_e^{t-1}, \varphi_i) > 0.5)), \end{aligned} \quad (14)$$

where $\alpha(t; \tilde{\mu})$ is as defined in Equation (9), $\hat{\mathcal{L}}_i(\theta_e^{t-1})$ is a loss function calculated with the predicted response $\mathcal{M}(\theta_e^t, \varphi_i; \omega)$ and the pseudo label $\mathbb{I}(\mathcal{M}(\theta_e^{t-1}, \varphi_i) > 0.5)$. The meta-learned gradient potentially provides better measure for the value of items than the original gradient via the adaptation term $\alpha(t; \tilde{\mu})$, which contains richer information through meta-optimization on massive historical data. In this sense, the meta-learned gradient embedding provides a suitable connection between item selection and cognitive state estimation. To be specific, when the CAT system adopts a vanilla estimator which performs multi-step gradient descent, the original gradient is a crude model influence approximation because of the mismatch to the actual update. In contrast, if we apply MCSE which performs single-step optimization, the meta-learned gradient embedding (Equation (14)) matches the update procedure well (Equation (8)). The elimination of mismatch motivates a better cooperation between the two modules.

4.4 Discussion

Having introduced the overall methodology, we delve into further explanation toward the proposed framework. Firstly, we give an intuitive interpretation of the meta-parameters modeled in MAMGE. Secondly, we conclude major differences and extension from our previous work, MAAT [8]. Thirdly, we highlight the major novel properties of MAMGE.

4.4.1 Meta-Knowledge Interpretation. Generally speaking, the meta-parameters modeled in MCSE, i.e., $\tilde{\theta}$ and $\tilde{\mu}$, are encoded with the *meta-knowledge* contained in the historical data, so that the original optimization process of the CDM presented in Equation (6) can be refined to Equation (8). Here we discuss what exactly such meta-knowledge is in the context of adaptive testing. As an intuitive metaphor, we consider a situation where a proficient home tutor judges a student's state through a series of tests to personalize the next study plan. In addition to the standalone performance of the specific student, the tutor naturally utilizes her *prior experience*, i.e., the students she has taught and the tests she has given. Then we make a correspondence between the prior experience owned by the home tutor and meta-knowledge encoded in meta-parameters.

(1) *Initialization term.* The meta-parameter $\tilde{\theta}$ used as the initialization term in Equation (8) can intuitively fit into the situation where the student has taken no test. Rather than set the initial study plan at a random difficulty level for the student, the tutor will probably specify a moderate level, which somehow accords to the overall ability distribution of her historical students, such as the average ability. Similarly, the meta-knowledge learned by $\tilde{\theta}$ can also be recognized as some distributional information about historical students. In this sense, under the assumption that the new student follows the same distribution, the meta-knowledge-enhanced initialization can outperform a random one.

(2) *Adaptation term.* The meta-parameter $\tilde{\mu}$ used as the adaptation term in Equation (8) can fit into the situation where the tutor fixes her judgment through prior experience. Consider the case where the student gets full marks in the first test. Taking no prior knowledge into consideration, the student's ability might be extremely overestimated, and the following personalized study plan for him would be overwhelmingly difficult. However, in practice, besides the standalone performance of the specific student, the teacher can refer to all his historical students who have taken the same test and then gives a moderate judgment as well as a reasonable plan. In correspondence, the meta-knowledge learned by $\tilde{\mu}$ is also such correction capability preventing the overfitting problem during parameter update.

Furthermore, we present case study in Section 5.2.4 which demonstrates a consistent behavioral pattern of MCSE influenced by the meta-parameters as discussed above.

4.4.2 Comparison with Previous Work. MAMGE is a natural and nontrivial extension of our previous MAAT framework [8] at the following two different levels.

(1) *Extension to the scope of study.* At a higher level, MAMGE considers adaptive testing as a systematic union of item selection and cognitive state estimation, while MAAT solely focuses on the item selection part. In this sense, MAMGE extends the scope of study on the adaptive testing problem, which is more practical and thus more effective (experimental comparison in Section 5).

(2) *Extension to the design of strategy.* Even within the scope of the item selection part, MAMGE extends to the methodological design by further exploring the role of gradients in MAMGE. Generally, both MAMGE and MAAT represent informativeness via gradient length, and the major difference lies in the representativeness heuristic. Specifically, to obtain item similarity, MAMGE utilizes the directions of gradient embeddings, while MAAT uses the associated knowledge points. We address two advantages of the design choice in MAMGE. Firstly, MAMGE minimally relies on external information of data and is thus more generally applicable than MAAT which requires knowledge of items annotated by experts. Secondly, since the gradient embeddings are computed w.r.t. the underlying CDM, gradient-based heuristics can intrinsically leverage the expressiveness of the CDM. For example, some recent CDMs powered by deep learning techniques are capable of extracting complex relationship among knowledge points from knowledge graphs [15, 45]. Nevertheless, the two designs have internal consistency rather than conflict. Consider a CDM whose cognitive state parameters are strongly correlated to knowledge, e.g., a vector representing

ability level on every knowledge point. Then, for an item associated with specific knowledge points, the direction of its gradient embeddings is believed to convey consistent information with the associated knowledge annotations.

4.4.3 Highlighted Properties. The key properties of MAMGE can be summarized from the following three aspects.

(1) *A model-agnostic framework.* In the previous discussion, the detail of the underlying CDM $\mathcal{M}(\theta, \varphi)$ is hidden, indicating that the proposed MAMGE framework can be applied to a wide range of CDMs. To be more specific, for both MCSE and GEIS, the only assumption is that the loss function is smooth enough in θ to support gradient-based operations. Such assumption is satisfied for most existing models, especially the recent deep learning-based ones [53]. Therefore, MAMGE is a flexible model-agnostic CAT framework.

(2) *A hybrid of heuristic-based and data-driven approaches.* Previous CAT approaches follow either a heuristic-based or data-driven fashion. Nevertheless, the former lacks effective data utilization while the latter suffers from data biases. In practice, it is hard to decide which is better before applying the approach. We argue that MAMGE has a more robust performance because it can be recognized as a hybrid of heuristic-based and data-driven solutions. On the one hand, the MCSE module makes an effective use of historical data by meta-learning the gradient embeddings. On the other hand, the GEIS module comprises of well-designed heuristics, making MAMGE able to perform at least similar to MAAT in case the meta-learned gradient embeddings degenerate to vanilla gradients.

(3) *A good connection of estimation and selection.* In existing CAT systems, the item selection module and the cognitive state estimation module are designed independently, which principally limits the testing effectiveness. On the one hand, the major consideration in item selection is how to quantify the potential information contained in each item. On the other hand, cognitive state estimation is exactly the way that an item conveys its information to the underlying CDM. In this sense, there is an intrinsic motivation to connect the two modules unifiedly. MAMGE realizes an ingenious connection using the meta-learned gradient embeddings which simultaneously serve as the selection indicator for GEIS as well as the estimation update for MCSE.

5 Experiments

5.1 Experimental Settings

5.1.1 Data Description and Evaluation Protocols. We evaluate the effectiveness of our framework with four datasets consisting of real-world students' interactions which are collected from different sources and have different sizes. The description of these datasets is listed as follows:

- *ASSIST*: *ASSISTments 2017* is a widely used educational dataset containing students' practice on mathematics with the ASSISTments system. *Junyi*: *Junyi Academy Math Practicing Log* is one of the largest publicly available student interaction datasets collected by the Junyi e-learning platform.
- *EXAM*: The *EXAM* dataset was supplied by iFLYTEK Co., Ltd., collected from an intelligent tutoring system. Different from ASSIST and Junyi, the interaction data in EXAM are collected when students took real math exams instead of practicing.
- *Examination for the certificate of proficiency in English (ECPE)*: The ECPE dataset is from a standard English test developed and scored by the English Language Institute of the University of Michigan.

Table 1. Statistics of the Preprocessed Datasets

Dataset	ASSIST	Junyi	EXAM	ECPE
#Students	1,670	29,301	3,750	2,922
#Items	1,960	701	1,179	28
#Logs	355,376	1,423,438	158,178	81,816
#Avg. logs/student	212.80	48.58	42.18	28.00
#Avg. logs/item	181.31	2030.58	143.41	2922.00
Label distribution	43.18%	76.67%	66.87%	71.46%

Avg., average.

For the reliability of the experimental results, we apply the following data preprocessing for all the datasets. First, if a student has multiple records on the same item, we remove the duplicates by keeping only the earliest one. Then we filter out the items answered by less than 10 students and the students that answered less than 20 items.

The analytic statistics presented in Table 1 indicate that the chosen four datasets have covered a variety of attributes. Firstly, the datasets are at a wide range of scales. For example, Junyi collects over a million logs from thousands of online students, while ECPE takes a much smaller-scale offline collection. Secondly, the datasets have different data density coming up with the average number of logs per student/item. In the experiments, higher per-item density (like Junyi and ECPE) means less variance in item calibration, i.e., how accurate the item representation can be. Higher per-student density (like ASSIST) means greater freedom in item selection validation, because experimentally we could only select those items with ground truths. In particular, for ECPE, each student has a log of answering *every* item (i.e., #Average logs/student = #Items) so that the limitation of item ground truth is completely removed, which is the reason we choose ECPE in our case study. Last but not least, the datasets have various label distributions (i.e., the ratio of the number of correct answers to the total number of logs) ranging from 43.18% to 76.67%. The wideness in dataset attributes can help support a more comprehensive and convincing experimental validation.

For each dataset, we randomly divide the students by 60%, 20%, and 20% for training, validation, and testing, respectively. The training partition plays the role of the historical examinees introduced in Section 3, i.e., $\mathcal{E}_{train}$, which is assumed known to the CAT system. Correspondingly, the testing partition acts as the new examinees, i.e., $\mathcal{E}_{test}$, to evaluate the effectiveness of the adaptive testing methods. In addition, the validation partition, which is similar to the testing partition, is used to perform early stopping and tune the hyperparameters for every method.

During testing, we use different methods to select a personalized set of items for each examinee with the same testing length (i.e., the same number of items). Besides, we practically limit our selection to those items whose actual response has been recorded in the testing data. Thereafter, as discussed in Section 3, we compare the testing effectiveness (i.e., $\psi(\theta^T)$) of different methods by predicting her/his performance on an evaluation item set. In this article, we use *all* the items whose actual responses have been recorded as the evaluation set. Specifically, we adopt two common metrics, ACC and the area under the receiver operating characteristics curve (AUC), to quantify $\psi(\theta^T)$.

5.1.2 Baseline Approaches. To demonstrate the effectiveness of MAMGE, we compare it with the selected baseline approaches listed as follows, including a basic random method as a scratch line, two classic IRT-based methods in the CAT literature [10, 31], a commonly used active learning approach [41], and two recently developed state-of-the-art methods [8, 16].

- *RANDOM*: This basic *Random* approach arbitrarily selects the next item for all examinees, which quantifies the improvement of the other approaches.
- **Maximum Fisher Information (MFI)**: *MFI* [9, 31] is the most popular selection strategy which measures the information of questions with the Fisher information function and is designated to the IRT model.
- *KLI*: *KLI* [10] is a global information heuristic that measures the informativeness with the aid of Kullback-Leibler divergence, designated to the IRT model as well.
- **Uncertainty Sampling (US)**: *US* [41] is a commonly used active learning method which selects the most uncertain samples for the current model to accelerate the learning procedure of the model. In our experiments, we use the widely adopted entropy function as the measure of uncertainty.
- *BOBCAT*: *BOBCAT* [16] automatically learns a data-driven item selection algorithm from training data within a bilevel optimization framework.
- *MAAT*: *MAAT* [8], making the first step toward a model-agnostic CAT system, guarantees the selected items with both quality and diversity through active learning techniques.

5.1.3 Underlying CDMs. To evaluate the model-agnostic property of the proposed framework, we apply all the above baselines as well as MAMGE on two different underlying CDMs if possible (note that the classical MFI and KLI methods can only apply to IRT). The one is the IRT [12] model as presented in Equation (1), which is the most commonly used CDM in CAT. The other is a neural network-based model used in [16], which we dub as *NN*. In specific, *NN* is implemented as a two-layer fully connected network with ReLU activation and 20% dropout rate, which consists of an 8-dimensional input layer, 64 hidden nodes, and a final sigmoid output layer that directly outputs the performance prediction for each item.

5.1.4 Parameter Settings. For all the baseline approaches, the hyperparameters are tuned for the best performance on the validation datasets. For our MAMGE framework, the hyperparameters are set as follows. In the MCSE module, we set $\gamma = 1$ for balancing the base part and dynamic part of the meta-learned adaptation term (Equation (9)). For the dynamic part, the LSTM is implemented with two recurrent layers. During meta training, the learning rates of both item-feature parameters and meta-parameters are set to 0.001. In the GEIS module, the balancing parameter K_C for the two-stage item selection (Equation (5)) is set to 1 when working with the IRT model, and 5 when working with the *NN* model. That means, for IRT models, we solely use the informativeness based heuristic, because the cognitive state parameters along with their gradients are only unidimensional in IRT, making the representativeness based heuristic useless. We set the maximum test length as 10, which is representative enough for typical CAT. Further discussion on the test length will be taken later. All the experiments are conducted on a Linux server with two 2.30 GHz Intel(R) Xeon(R) Gold 5118 CPUs and one 11G GTX 1080ti GPU.

5.2 Results and Analysis

5.2.1 Overall Performance. In Table 2, we show the average overall performance comparison on the ACC metric after finishing the whole test (i.e., when reaching the maximum test length); in Table 3, we do the same using the AUC metric. Outperforming results are highlighted in bold in each table. We can clearly see that, despite the different underlying models and evaluation metrics, MAMGE consistently achieves significant improvements compared with all the state-of-the-art baselines. All the four datasets show similar results. It proves that MAMGE can effectively achieve an MAAT system. In addition, there are some further observations worth discussing.

Firstly, we notice that the *NN* model outperforms the IRT model in most cases, yet with some exceptions. This observation indicates two points. On the one hand, there indeed exist situations

Table 2. The Overall Performance of MAMGE with ACC Metric on Four Real-World Datasets Compared with Baselines

	ACC (%) on IRT Model				ACC (%) on NN Model			
Dataset	ASSIST	Junyi	EXAM	ECPE	ASSIST	Junyi	EXAM	ECPE
RANDOM	68.59	81.64	75.70	74.56	69.65	80.90	76.21	75.53
MFI [31]	69.51	82.61	77.48	76.86	-	-	-	-
KLI [10]	69.30	79.39	77.12	76.88	-	-	-	-
US [41]	69.28	82.59	77.58	76.82	70.45	83.17	77.58	77.97
BOBCAT [16]	69.01	81.91	77.32	76.33	70.41	82.43	77.56	76.09
MAAT [8]	69.87	82.65	77.73	76.79	70.54	82.19	76.77	77.31
MAMGE	70.67	82.67	78.39	77.44	71.08	84.81	81.15	81.90

Outperforming results are shown in bold.

Table 3. The Overall Performance of MAMGE with AUC Metric on Four Real-World Datasets Compared with Baselines

	AUC Score (%) on IRT Model				AUC Score (%) on NN Model			
Dataset	ASSIST	Junyi	EXAM	ECPE	ASSIST	Junyi	EXAM	ECPE
RANDOM	74.19	80.97	80.58	76.38	75.45	79.43	81.78	77.36
MFI [31]	75.52	82.73	81.19	77.29	-	-	-	-
KLI [10]	74.29	81.53	81.26	77.36	-	-	-	-
US [41]	74.63	82.69	81.50	77.32	75.97	80.06	81.78	78.53
BOBCAT [16]	75.16	82.77	82.46	76.61	75.87	81.28	82.30	77.75
MAAT [8]	75.52	82.77	81.50	77.49	75.70	81.02	83.07	79.96
MAMGE	76.04	83.02	83.08	77.64	77.31	84.19	84.85	81.70

Outperforming results are shown in bold.

where some CDMs are somewhat more “powerful” than others. On the other hand, simply keeping designing new algorithms for those more powerful CDMs could not solve the problem completely, because it is impossible to fully assert how a CDM will perform on a dataset. Therefore, it is necessary for a CAT framework to be model-agnostic due to the various performance of different CDMs on different datasets. In this sense, MFI and KLI, proven to be classical and efficient IRT-based approaches though, are difficult to extend to a wider range of CDMs, including those more powerful ones such as NNs. In contrast, MAMGE consistently performs well with both underlying CDMs. In the meantime, we find that MAMGE makes a larger margin of improvement with NN model than that with IRT model, which indicates that MAMGE is fairly suitable for more complex and powerful CDMs.

Secondly, by observing the results of US which is a simple strategy applicable to nearly all the models, we find that it is nontrivial for a model-agnostic method to achieve outstanding performance at the same time. In fact, due to the inaccessibility of the detail of the underlying model, there is additional demand for sophisticated designs in the adaptive testing framework. In comparison, MAMGE simultaneously keeps flexibility and effectiveness with carefully designed modules for both item selection and cognitive state estimation.

Thirdly, compared with the state-of-the-art work BOBCAT, our method shows more robustness and stability. Especially, we observe that BOBCAT performs trivially on ECPE, which lacks a large

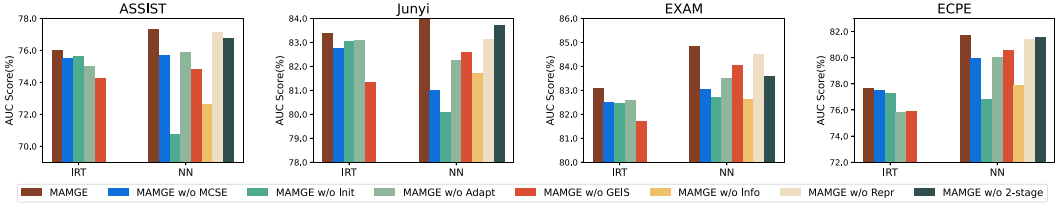


Fig. 5. The ablation studies of MAMGE with AUC metric on four datasets.

quantity of interaction data compared to the other datasets. The key point behind is that BOBCAT is thoroughly data-driven, which automatically learns the whole item selection module as well as the parameter initialization from historical data. Therefore, BOBCAT is prone to suffer from biases or scarcity in historical data, which is also mentioned in the original article [16]. Besides, we empirically recognize a higher difficulty in training and parameter tuning for BOBCAT during our experiments. In contrast, MAMGE has a better compromise between data-driven and heuristic-based ideas as discussed in Section 4.4.

Last but not least, looking at the results of MAAT which MAMGE extends from, we can notice that, in accord with the basic idea of this article, better cognitive state estimation for CAT can bring a great improvement. As discussed in Section 4.4, we argue that the advantages come from not only the estimator itself but also the reasonable combination of cognitive state estimation and item selection.

5.2.2 Ablation Study. To further verify the effectiveness of different parts, we compare MAMGE with its variants. In addition to validating MCSE and GEIS wholly, we conduct further fine-grained validation for detailed parts inside each of the two modules. The compared variants are listed as follows:

- *MAMGE w/o MCSE*: To verify the effectiveness of MCSE, we only apply a vanilla estimator presented as Equation (6) in cognitive state estimation.
- *MAMGE w/o Init*: Inside the MCSE module, to verify the effectiveness of the meta-learned initialization term $\tilde{\theta}$ (Equation (8)), we only apply random initialization in cognitive state estimation.
- *MAMGE w/o Dyna*: To verify the effectiveness of the dynamic part of the meta-learned adaptation term α_{dyna}^t in MCSE module, we disable it by setting $\gamma = 0$ in Equation (9).
- *MAMGE w/o GEIS*: To verify the effectiveness of GEIS, we use a random strategy for item selection.
- *MAMGE w/o Info*: Inside the GEIS module, to verify the effectiveness of the informativeness-based heuristic, we apply only the representativeness-based heuristic by setting $K_C = |\mathcal{I}_U|$ during the two-stage item selection.
- *MAMGE w/o Repr*: To verify the rationality of the representativeness based heuristic in GEIS module, we apply only the informativeness based heuristic by setting $K_C = 1$ during the two-stage item selection.
- *MAMGE w/o Two-stage*: To verify the effectiveness of two-stage item selection, we replace it with weighted sum to combine the informativeness-based heuristic and the representativeness-based heuristic. Formally, we replace Equation (5) with $i_t = \arg \max_{A \in \mathcal{I}_U^{t-1}} \text{Info}(i) + w * \text{Repr}(i)$, where $w \in (0, +\infty)$ represents the sum weight. In our ablation study, we tune w to 0.1 with hyper-parameter optimization on validation sets.

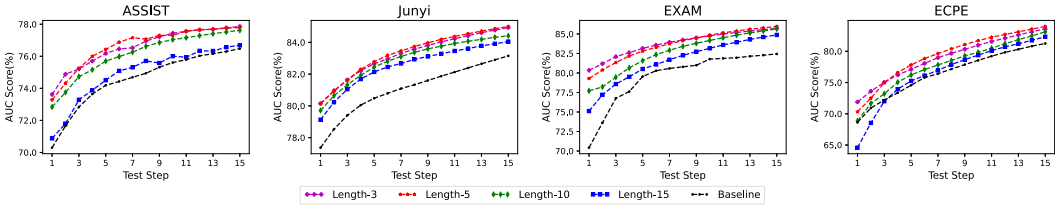


Fig. 6. The analysis on the effect of the test length with AUC metric on four datasets.

Figure 5 depicts the comparison results on the four datasets with the two underlying CDMs. Note that for IRT, since the representativeness based heuristic is disabled, three of the variants, *MAMGE w/o Info*, *MAMGE w/o Repr*, and *MAMGE w/o Two-stage*, are ignored. The overall results clearly show the effectiveness of the studied parts in MAMGE. Especially, there are several observations worth noting.

Firstly, from the comparison results of *MAMGE w/o MCSE* and *MAMGE w/o GEIS* with IRT model, we can find that the cognitive state estimation module (i.e., MCSE) plays a less important role than the item selection module (i.e., GEIS). The reason is that cognitive state estimation in IRT is much simpler due to its one-dimensional representation. Under this situation, the meta-learned estimator cannot be shown to best advantage. In contrast, for NN model, the MCSE module has a critical impact on the performance because of its complex representation with neural layers. Moreover, it also well explains why MAMGE makes a larger margin of improvement with NN model than that with IRT model.

Secondly, by looking into the results of *MAMGE w/o MCSE*, *MAMGE w/o Init* and *MAMGE w/o Dyna* with NN model, we observe that: (1) The initialization term is fairly important for MCSE when estimating complex cognitive state parameters, without which the meta-learned estimator will even perform worse than vanilla estimators; and (2) The modeling of the dynamics in cognitive state estimation is effective, since the performance of MCSE significantly degrades without the dynamic part of the adaptation term.

Thirdly, the results of *MAMGE w/o GEIS*, *MAMGE w/o Info*, and *MAMGE w/o Repr* make a good confirmation for our understanding of the item selection heuristics. On the one hand, the informativeness based heuristic is the central consideration during item selection, without which the strategy performs even worse than random selections. On the other hand, the representativeness based heuristic plays a role in regularization to prevent the GEIS module from selecting those out-of-distribution ones in the item pool.

Finally, *MAMGE w/o 2-stage* illustrates an alternative way of combining the informativeness-based heuristic and the representativeness-based heuristic using weighted sum instead of two-stage sequential selection. Note that the combination method is more of a design choice rather than our contribution, and the MAMGE framework is completely flexible to use other methods. Nevertheless, the results show that the weighted-sum-based selection has competitive but slightly lower performance than two-stage selection, which validates that the latter is a rational choice.

5.2.3 Analysis on Test Length. This subsection discusses the test length, i.e., the number of steps in adaptive testing. In MAMGE, the effect of test length is two-fold: (1) the test length of the sampled task for meta-training (Algorithm 2) and (2) the test length of CAT for real new examinees (Algorithm 1). In the previous experiments, the two concepts are treated the same. Furthermore, the length of adaptive tests is assumed to be fixed, which is sometimes impractical. For instance, the tests may adopt a stopping rule [11] resulting in variable lengths. Along this line, three questions

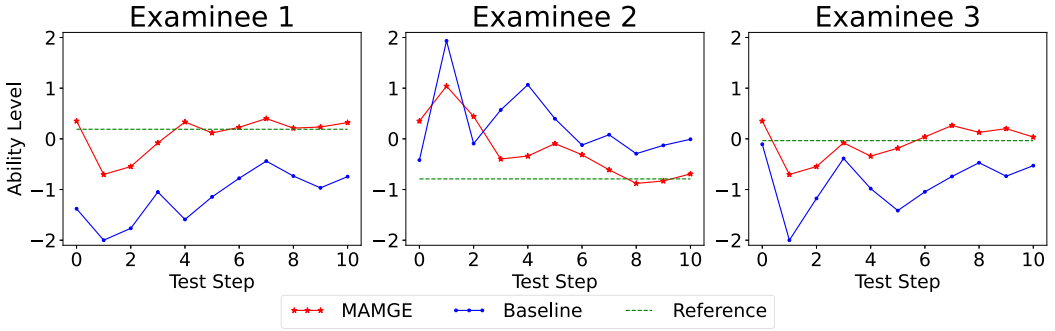


Fig. 7. Visualization of the cognitive state estimation procedure of the adaptive tests for three examinees in the ECPE dataset.

naturally arise: (1) Can we meta-train with a test length different from the one when conducting adaptive testing? (2) Can we perform variable-length adaptive tests with the trained framework? (3) How does the meta-training length influence the performance of the framework?

To answer the questions, we meta-train the MCSE module with various lengths of support sets (3, 5, 10, and 15), and then conduct adaptive tests with various numbers of steps (from 1 to 15). Figure 6 illustrates the results on four datasets with NN model, where each line in the figure depicts the AUC metric of MAMGE meta-trained with a certain length, except that the US method (the black line) acts as a baseline. By comparing the results, we have the following observations. First, instead of achieving good performance only at the test step the same with the meta-training length, each line has a normal uptrend, i.e., the performance keeps a healthy growth with the test steps, which verifies the feasibility to meta-train MAMGE with a flexible test length different from the one in adaptive testing. Second, almost all the lines are significantly beyond the baseline (except initial steps of *Length-15* on ECPE), indicating that our framework is robust and performs consistently well in the context of various test lengths. Last but not least, by carefully comparing the different lines in Figure 6, we can find that the meta-training length influences the trade-off between *initialization* and *adaptation* of MAMGE in effect. The shorter the meta-training length, the better performance at initial steps. For instance, at the first test step on all the datasets, *Length-3* has the best performance while *Length-15* is the worst one. Nevertheless, with a longer the meta-training length, MAMGE is able to adapt faster, which potentially leads to better performance in the long range. For example, on ECPE, *Length-5* outperforms *Length-3* after 3 test steps. In general, in Figure 6, the framework meta-trained with *Length-5* has the best overall performance across test steps among the four datasets. Understanding this trade-off can help achieve high robustness and performance in practice.

5.2.4 Case Study. To illustrate how the MCSE module in MAMGE works, we visualize the cognitive state estimation procedure on some representative cases. Specifically, Figure 7 shows three examinees' cognitive states represented by IRT model, which are scalar values reflecting their ability level. The red lines depict the values estimated by MAMGE, while the blue lines depict the MFI [31] method which has a vanilla estimator. For a more intuitive explanation, the dashed green lines represents the reference ability levels for each examinee, which are calculated with all the records of the examinees. First of all, we can see that MAMGE initializes the ability level to the same value for all the examinees, which is meta-learned from historical data so that later estimation can adapt fast. In contrast, the baseline randomly samples an initial value, which sometimes results in suboptimal estimation. In the case of Examinee 1, the baseline begins with an extremely low level and the examinee happens to answer wrongly to the first item. Although the examinee performs

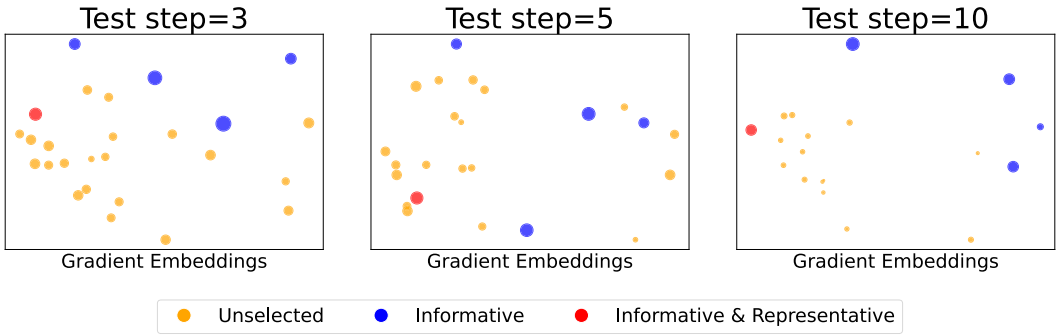


Fig. 8. Visualization of the item selection procedure of the adaptive test for an examinee in the ECPE dataset.

better on the later items, the underestimation can hardly be corrected. In addition to initialization, MAMGE also shows robustness against overfitting. In the case of Examinee 2, the initial value of the baseline is closer to the reference value than that of MAMGE. At the first test step, the baseline gets overfitted and overestimating the ability level to extremely high, but differently, MAMGE takes a tender step that makes it adapt well in the following steps. Finally, looking into the curves in all the three cases, we find that MAMGE tends to take a larger step size to adapt fast in the earlier stage, and a smaller step size to fine-tune in the later stage. This observation verifies the effectiveness of the dynamic part of the adaptation term in MCSE (Equation (10)). In effect, the step size can be dynamically adjusted by the recurrent layers as the adaptive test goes on. The above behavior patterns of MCSE are exactly in correspondence to the intuitive interpretation of meta-parameters discussed in Section 4.4.

To illustrate how the GEIS module in MAMGE works, we also visualize the item selection procedure. Figure 8 shows part of the item selection procedure (test step $t = 3, 5, 10$) for an examinee in ECPE dataset with NN model. In the figure, each circle represents the gradient embedding of an unselected item which is projected into a two-dimensional space through principle component analysis. The radius of a circle is proportional to the norm of the gradient embedding (before dimension reduction). Red and blue circles are those K_C items selected based on informativeness, among which the red circle is the final choice by the representativeness based heuristic. The illustration verifies that, besides informativeness, MAMGE takes representativeness of items into consideration as well. We note that the final choice of MAMGE is not always the one with maximum informativeness (i.e., the circle size). Instead, the representativeness-based heuristic effectively keeps away from those out-of-distribution items and prefers the one which represents the underlying distribution better. Another interesting observation is that the average radius of the circles in the figure becomes smaller from left (test step = 3) to right (test step = 10), indicating smaller step sizes. This fact again verifies that MAMGE tends to dynamically degrade its step size for fine-tuning.

6 Conclusion and Future Work

In this article, we presented a focused study on CAT for intelligent education systems, i.e., to improve the personalization and efficiency of testing by administrating automated item selection and cognitive state estimation for examinees. Specifically, we proposed an MAAT framework via MAMGE to address the challenges in item selection and cognitive state estimation with a unified manner. Firstly, we designed an MCSE to automatically control the estimation process by learning to learn a proper initialization, update direction as well as step size. Then, we designed a GEIS which uses the pseudo meta-learned gradients provided by MCSE as embeddings to represent items and selects the best ones that are simultaneously informative and representative. Both MCSE and

GEIS are inherently model-agnostic and have an intrinsic connection via the concept of gradient embeddings. Finally, we conducted extensive experiments to verify effectiveness as well as the model-agnostic property of MAMGE on four real-world datasets with different underlying CDMs. In our future work, we will continue to explore adaptive testing methods with a focus on the scalability and explainability. In addition, we also notice rising opportunities for extending the application scope of CAT. Besides the field of education, there exist more intelligent systems needing an efficient and effective user cognitive modeling component, e.g., knowledge tracing systems [43], psychology-informed or cross-domain recommendation systems [25, 54, 60], and social behavior analysis [57]. We hope this work can give some insights for future research.

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